Candidate Seat no…………

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (EE) Examination

May 2018

EE307 Electrical Power System III

Date: 03/05/2018, Thursday **Time: 10.00 a.m. to 1.00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1 A Choose the correct option. **[05]**

I. The arc voltage in a circuit breaker is
(a) in the phase with the arc current. (b) lagging the arc current by 90°.
(c) leading the arc current by 90°. (d) lagging the arc current by 180°.

II. In a circuit breaker, the active recovery voltage depends upon
(a) circuit conditions. (b) power factor.
(c) armature reaction. (d) all of these.

III. In a circuit breaker the current that exists at the instant of contact separation is called the... current.
(a) restriking (b) breaking (c) arc (d) recovery

IV. Which of the following statements is not correct?
(a) SF6 is non-toxic and non-inflammable gas.
(b) SF6 has very high dielectric strength roughly 24 times of that of air.
(c) SF6 is yellow in colour.
(d) SF6 is about 100 times more effective than air in arc extinction.
(e) SF6 has density 5 times that of air at 20°C.

V. For magnetic blow out of arc the magnetic field is produced
(a) in the load circuit. (b) at right angles to the axis of the arc.
(c) in line with the axis of the arc. (d) any of the above.

B Identify whether the following statements are TRUE or FALSE. **[02]**

I. RRRV of a system is expressed in kV/km.
II. SF6 gas is not an efficient quenching medium because of its electro-negativity.

Q – 2 **Answer any THREE** **[15]**

A With neat and clean figures, waveforms and relevant data explain the current chopping phenomena in circuit breaker. Also explain the adverse impact of current chopping on circuit breaker performance.

B With neat and clean, necessary circuit diagram and waveforms, justify that the circuit breaker which is used for capacitor bank switching should be re-strike free.

C Derive the expression of "Critical Damping Resistance" used in resistance switching.

D For a 400 kV, 50 Hz system, the inductive reactance and capacitance up to the location of the circuit breaker is 20 ohms and 0.025 μF, respectively. Calculate the following. (Neglect first pole to clear factor):

(a) Peak restriking voltage across the circuit breaker.

(b) Frequency of restriking voltage transient.

(c) Average rate of restriking voltage up to the peak restriking voltage.

(d) Maximum R.R.R.V.

Q – 3. A With neat diagram explain the construction and working of Air Break Circuit Breaker. **[06]**

B Explain the difference causes due to which ionization and deionization of gases take place. **[07]**

OR

B With neat diagram explain the construction and working of Vacuum Circuit Breaker. **[07]**

SECTION – II

Q - 4 Choose the correct option. **[07]**

I. For the power system shown in figure 1, single line to ground fault occurs at bus D, the fault current magnitude between transformer and bus D is 6.00 pu. The possible magnitude of zero sequence component between generator and transformer in pu is….

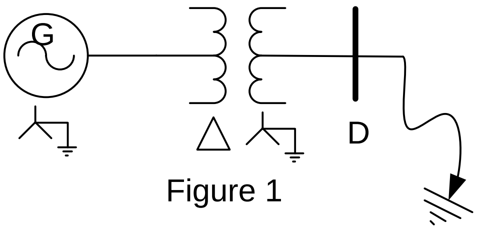


Figure 1

(a) 2.00 (b) $2*\sqrt{3}$ (c) $2/\sqrt{3}$ (d) 0

II. The sequence components of fault current are as under.

$I_{b1} = 6.05\angle 220^0\ pu;\ I_{a2} = 0;\ I_{c0} = not\ known$

The possible fault is…

(a) L-G (b) L-L (c) L-L-G (d) L-L-L

III. For the power system shown in figure 2, for which combination of transformer winding connection, the fault current for L-G fault will flow.

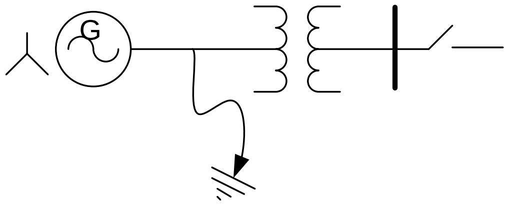
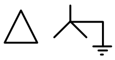
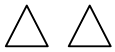
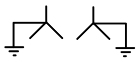
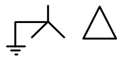


Figure 2

(a)  (b)  (c)  (d) 

IV. The power invariance theorem says that...
- (a) In all sequence network power remains same.
- (b) zero sequence network doesn't consume any power.
- (c) Power consume as per phase quantity is three times power consumes as per positive sequence quantity.
- (d) Power consume as per phase quantity is equal to sum of the power consume in positive, negative and zero sequence quantity.

V. For a particular fault, $I_{a0} = 3.72\angle 90°,\ I_{b1} = 5.66\angle 150°\ and\ I_{c2} = 1.94\angle -30°$. possible fault is____

(a) L-G (b) L-L (c) L-L-G (d) L-L-L

VI. The pu impedance of transformer is…
- (a) Higher value referred to HV side
- (b) Lower value referred to HV side
- (c) same value referred to any side
- (d) always 1.00 pu.

VII. In a star-connected system with neutral grounding, zero-sequence currents are
- (a) zero.
- (b) Phasor sum of phase currents.
- (c) three times Phasor sum of phase currents.
- (d) one third times Phasor sum of phase currents.

Q – 5. A A transmission line of inductance 0.095 H and resistance 8 Ω is suddenly short-circuited at the far end. Find the instantaneous value of AC current and DC offset current for following conditions. **[05]**

i. 0.06 second after fault, if sending end voltage at instance of fault is v=200sin(100πt+75º)

ii. 0.03 second after fault, if sending end voltage at instance of fault is v=200sin(100πt+210º)

B Draw the sub-transient, transient and steady state model of synchronous machine. Also draw the waveform of symmetrical fault considering the sub-transient, transient and steady state period. **[02]**

C A part of power system is shown in figure 3. Both motors are running at their 75% of full load capacity at unity power factor with their terminal voltage of 10.5 KV. At that time a three phase fault occurs at the generator busbar. Find the total fault current. **Take base as per generator rating.** **[07]**

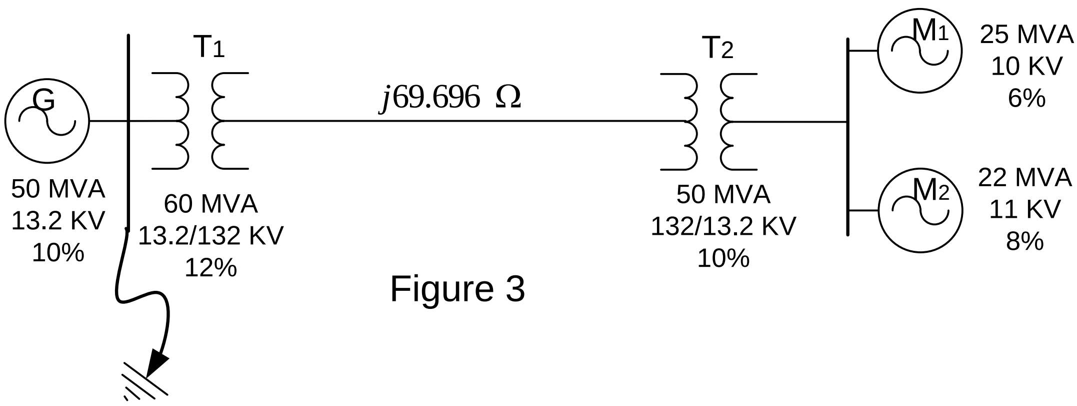


Figure 3

OR

Q – 5. A Derive the sequences impedances of transmission line. **[07]**

B A power station has two alternators of rating 5000 kVA, 8% reactance each and one alternator of 7500 kVA, 7.5% reactance as shown in figure 4. The circuit breakers are rated 300 MVA. The system is to be extended by taking supply from a grid via transformer of 10000 kVA rating having a reactance of 7.5%. If the same Circuit Breakers are to be used, find the necessary reactance to be placed between the bus-bar and the transformer to protect the switch gear. The voltage of the system at the bus-bar is 3300 volts. (**Take base KVA= 10000 KVA**) **[07]**

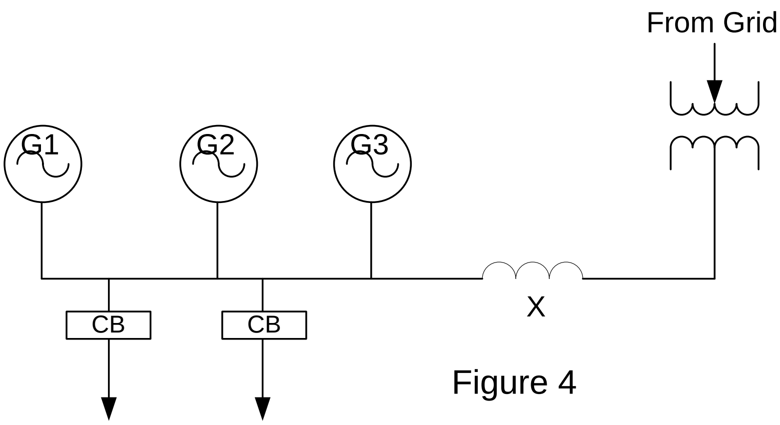


Figure 4

Q – 6. A Using appropriate interconnection of sequence networks, derive the equation for a line to line fault in a power system with a fault impedance of Z_f. **[07]**

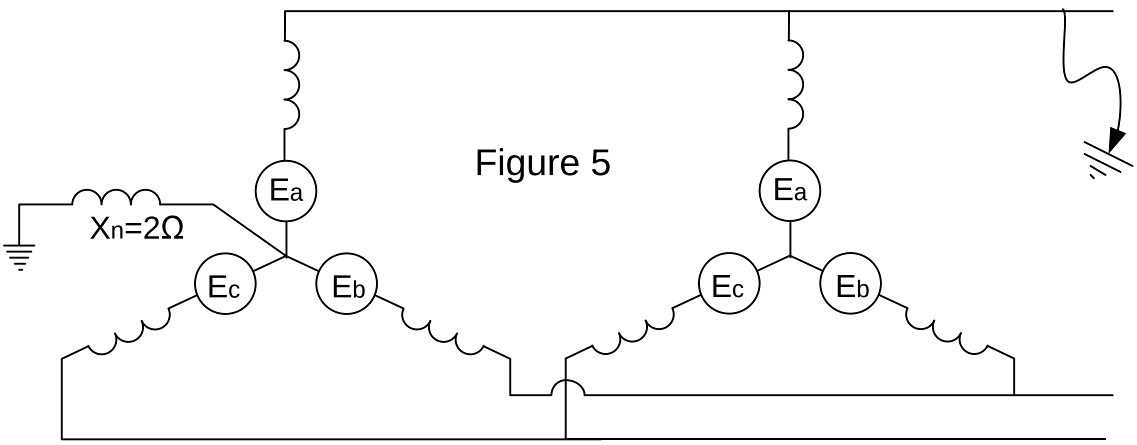
B Two 3-phase, 25 MVA, 11 KV, 50 Hz alternators operate in parallel as shown in figure 5. Each alternator have reactances of X_1=20%, X_2=15% and X_0=10%. The star point of one of the alternator is isolated and that of other is grounded through an inductor of 2.0 Ω. A L-G fault occurs at the terminal of one of the alternator, calculate.. **[07]**

(i) Fault current in ampere through grounding inductor.

(ii) Voltage across grounding inductor.

(iii) Fault current from all the phases of each alternator.

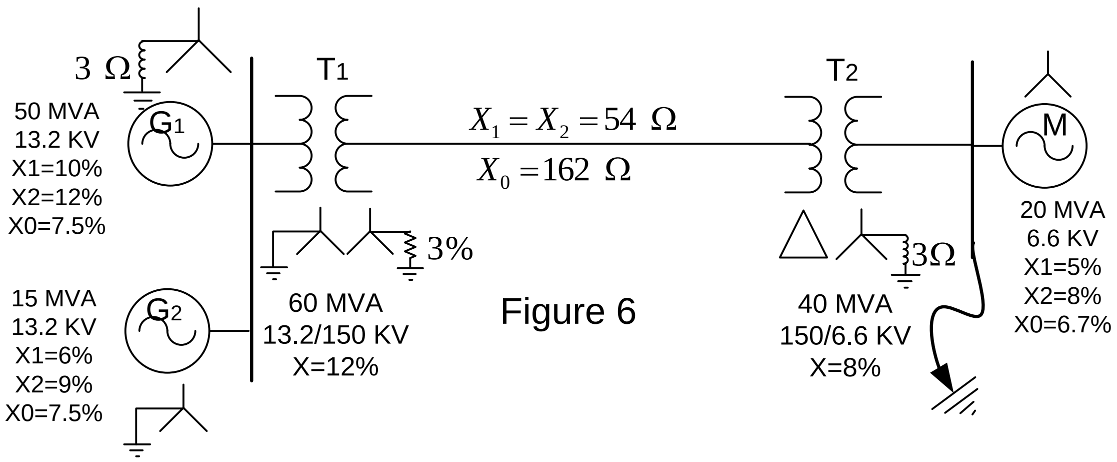
(**Take base as per alternator Rating.)**



OR

Q – 6. A Using appropriate interconnection of sequence networks, derive the equation for a Double line to ground fault in a power system with a fault impedance of Z_f. **[07]**

B A part of power system is shown in figure 6. Find the value of LL fault current in ampere when fault occurs at the terminal of motor as shown in figure. **Take base as per generator-1 rating.** **[07]**



***